NUMERICAL COMPUTING II

Assignment III

Can We Make Numerical Differentiation More Accurate? Investigation of Richardson Extrapolation

1. Aim

The aim of this assignment is to investigate whether Richardson Extrapolation can improve the accuracy of numerical differentiation. Central Difference and Richardson Extrapolation are implemented and compared using four test functions and different step sizes. The experiment also investigates truncation error, convergence rates, and floating-point round-off error.

2. Theory

Central Difference estimates a derivative using:

$$\mathbf{D(h) = [f(x + h) - f(x - h)] / (2h)}$$

Its leading truncation error is of order $O(h^2)$.

Richardson Extrapolation combines two Central Difference approximations, one using step size h and the other using h/2:

$$\mathbf{R(h) = [4D(h/2) - D(h)] / 3}$$

This combination cancels the leading $O(h^2)$ error term, giving a theoretical truncation error of order $O(h^4)$ for sufficiently smooth functions.

3. Experimental Setup

The program was implemented in Python using separate classes for Central Difference and Richardson Extrapolation. The evaluation point was x = 1. The step sizes ranged from 10^{-1} to 10^{-10}, allowing the behaviour of the methods to be studied as the step size became very small.

The test functions were:

1. $f(x) = e^x$
2. $f(x) = \sin x$
3. $f(x) = \cos x$
4. $f(x) = x^2 - 2x + 1$

The exact derivatives at x = 1 were e, cos(1), −sin(1), and 0, respectively. Absolute errors and log-log plots were used to compare the methods.

4. Algorithm

5. Select a test function and evaluation point x = 1.
6. Calculate the exact derivative.
7. Choose a step size h.
8. Calculate the Central Difference approximation D(h).

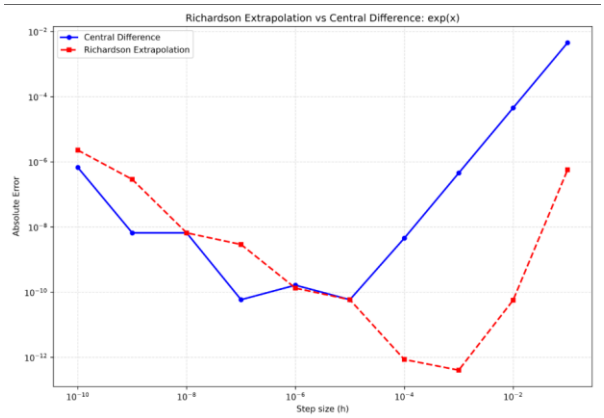
9. Calculate D(h/2).
10. Calculate the Richardson approximation using R(h) = [4D(h/2) − D(h)] / 3.
11. Calculate the absolute errors for both methods.
12. Repeat for all step sizes and test functions.
13. Print the results and plot step size against absolute error on logarithmic axes.
14. Compare observed convergence behaviour with the theoretical orders.

5. Results and Discussion

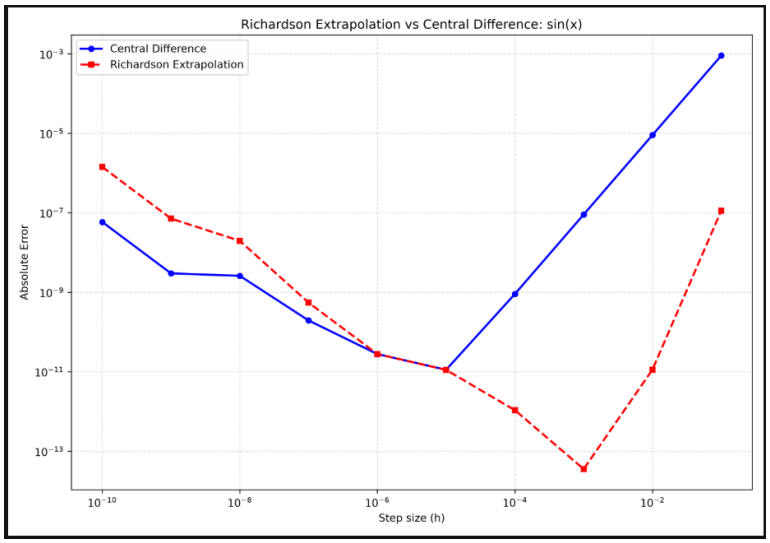
The results show that Richardson Extrapolation can substantially improve accuracy at suitable step sizes.

Function at h = 10^{-3}	**Central Difference error**	**Richardson error**
f(x) = e^x	4.53×10^{-7}	4.02×10^{-13}
f(x) = sin x	9.01×10^{-8}	3.54×10^{-14}
f(x) = cos x	1.40×10^{-7}	1.13×10^{-13}

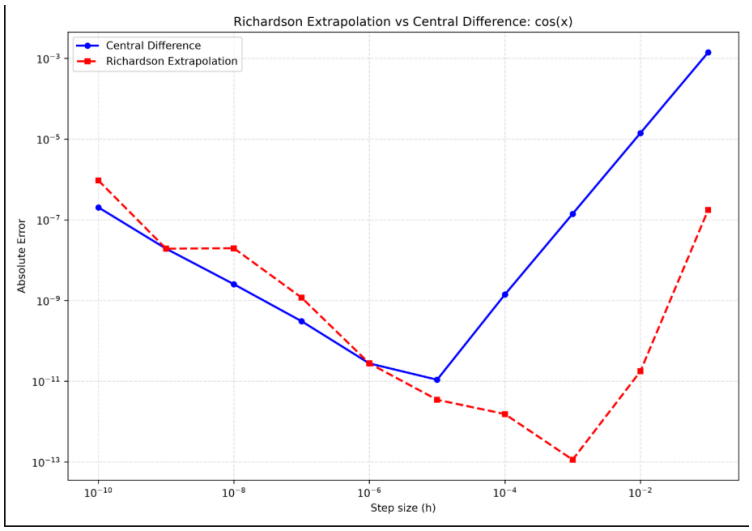
For f(x) = e^x, the Central Difference error at h = 10^{-3} was approximately 4.53×10^{-7}, while the Richardson error was approximately 4.02×10^{-13}.



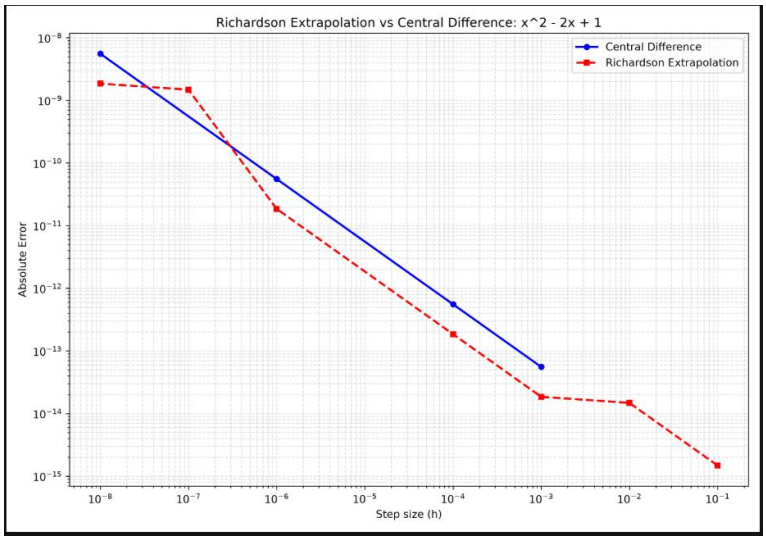
For f(x) = sin x, the Central Difference error at h = 10^{-3} was approximately 9.01×10^{-8}, while the Richardson error was approximately 3.54×10^{-14}.



For $f(x) = \cos x$, the Central Difference error at $h = 10^{-3}$ was approximately 1.40×10^{-7}, while the Richardson error was approximately 1.13×10^{-13}.



For the quadratic function $f(x) = x^2 - 2x + 1$, Central Difference is theoretically exact in exact arithmetic. In the computed results, both methods produced zero or very small errors because of floating-point arithmetic.



As the step size becomes extremely small, the errors for the exponential, sine, and cosine functions eventually increase. This occurs because floating-point round-off error becomes more significant. Therefore, a smaller step size does not always produce a more accurate result.

6. Answers to the Assignment Questions

Q1. Does Richardson always give a smaller error than Central Difference?

No. Richardson Extrapolation generally produces smaller errors when truncation error dominates, but it does not guarantee a smaller error for every step size and function. Round-off error can affect the extrapolated result, and Central Difference may already be exact or nearly exact for certain functions.

Q2. By approximately what factor does the error decrease when h is reduced by 10?

For Central Difference, the theoretical truncation error decreases by approximately $10^2 = 100$. For Richardson Extrapolation, it decreases by approximately $10^4 = 10{,}000$. These factors apply when truncation error dominates; actual results can differ because of round-off error.

Q3. Does Richardson continue to improve as h becomes very small?

No. Initially, reducing h improves accuracy, but after a suitable range the error stops decreasing and may increase. The calculations for the three smooth functions demonstrate this behaviour at very small step sizes.

Q4. Does the log-log slope support the theoretical prediction?

In the truncation-error-dominated region, the expected log-log slope is approximately 2 for Central Difference and 4 for Richardson Extrapolation. The plotted results should be interpreted over this region. At very small step sizes, round-off error causes deviations from these theoretical slopes.

Q5. Does Richardson achieve the expected $O(h^4)$ behaviour?

Richardson Extrapolation is theoretically fourth-order accurate for sufficiently smooth functions. The results for the exponential, sine, and cosine functions show substantial improvements consistent with the purpose of the method. However, the fourth-order trend cannot be expected to continue across every step size because round-off error eventually dominates.

Q6. What role does floating-point round-off error play?

Floating-point numbers have finite precision. When h becomes very small, $f(x + h)$ and $f(x - h)$ become very close. Subtracting nearly equal numbers can lose significant digits. Richardson Extrapolation combines multiple numerical approximations, so it can also be affected by round-off error. This explains why the error eventually increases instead of continuing to decrease.

7. Conclusion

The experiment demonstrates that Richardson Extrapolation can improve the accuracy of Central Difference by cancelling its leading truncation-error term. Central Difference has theoretical error order $O(h^2)$, whereas Richardson Extrapolation has theoretical error order $O(h^4)$ for sufficiently smooth functions.

Therefore, Richardson Extrapolation is an effective method for improving numerical differentiation, provided that the step size is selected carefully. The experiment also demonstrates the importance of balancing truncation error and floating-point round-off error.

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